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Smart Solar Tracking and Self-Cleaning Solar Panel System

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Course: Sensors

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Abstract

A smart solar tracking and self-cleaning solar panel prototype was designed and implemented for the Sensors course. The system was developed to improve solar panel performance by addressing two major factors that reduce energy output: non-optimal sun exposure and dust accumulation on the panel surface. The prototype uses four light-dependent resistors to perform dual-axis solar tracking, allowing the panel to rotate toward the direction of maximum light intensity. A TCRT5000 infrared reflective sensor is used to detect dust on the panel surface by monitoring changes in reflected infrared light. When dust is detected and environmental conditions are acceptable, a submersible water pump is activated through a relay module to spray water onto the panel. A load cell with an HX711 module is used to monitor the water mass in the tank, while a YF-S201 flow sensor is used to verify water flow through the tubing. A DHT22 humidity sensor and temperature sensor is also included to monitor environmental conditions that may affect the dust sensor readings. The system provides user feedback through an LCD. The final prototype was functional and successfully demonstrated solar tracking, dust detection, water-level monitoring, and automatic cleaning.

1. Introduction

Solar energy is one of the most widely used renewable energy sources because photovoltaic panels can convert sunlight directly into electrical energy. However, the output of a solar panel depends strongly on the amount of sunlight received by its surface. If the panel remains fixed while the sun moves throughout the day, the angle of incidence between the sunlight and the panel changes, which reduces the effective irradiance reaching the panel. In addition, dust, dirt, and small particles deposited on the panel surface can block part of the incoming light and further reduce the panel's performance.

This project was developed to address both issues in a single prototype. The system combines solar tracking and automatic surface cleaning. The tracking mechanism is used to keep the panel oriented toward the strongest light direction, while the cleaning mechanism is used to remove dust from the panel surface when a dust threshold is reached. The system was designed as a prototype rather than a full-scale solar installation. Therefore, a demonstration solar panel, small servo motors, a small water tank, a submersible pump, and Arduino-based sensor modules were used.

The project was implemented using an Arduino Uno as the main controller. The Arduino receives readings from the light sensors, dust sensor, water-level measurement system, flow sensor, humidity sensor, and temperature sensor. Based on these readings, the Arduino controls the servo motors, pump relay, and LCD. The result is an integrated sensor-based system that responds automatically to environmental and operating conditions.

2. Project Objective

The objective of this project was to design and implement a smart solar panel prototype that combines solar tracking and automatic self-cleaning in one integrated system. The project was developed to address two major factors that reduce the efficiency of photovoltaic panels:

1. Poor panel alignment with the sun which reduces sun exposure.
2. Dust accumulation on the panel surface which reduces the intensity of the received light.

The solar tracking part of the system was designed to continuously adjust the orientation of the panel according to the direction of maximum light intensity, allowing the panel to receive the highest possible amount of incident light throughout operation. This was achieved using four LDR sensors arranged on the moving panel, where the differences in light intensity between opposite sensors were used to control two servo motors and perform dual-axis movement. The self-cleaning part of the system was designed to detect dust on the panel surface using an infrared reflective sensor and activate a water-spraying mechanism when cleaning is required. In addition, the system was designed to check the available water level before activating the pump, verify water flow during cleaning, and display important system information such as water status, temperature, and humidity. Therefore, the overall aim of the project was not only to build a moving solar panel, but to create a complete sensor-based prototype capable of monitoring its environment, making decisions, and responding automatically to conditions that affect solar panel performance.

3. System Overview

The final system consists of a moving solar panel mounted on a 3D-printed dual-axis base. Four LDR sensors are mounted on the moving panel and separated by barriers. The

LDR readings are continuously compared to detect differences in light intensity. When one side of the panel receives more light than the opposite side, the corresponding servo motor is adjusted to reduce the difference and align the panel toward the optimal light direction.

The cleaning subsystem is based on a TCRT5000 infrared reflective sensor mounted on the panel. This sensor detects changes in the reflection from the panel surface. When dust accumulates on the surface, the amount of reflected infrared light changes. If the measured change exceeds a selected threshold, the system interprets the panel as dusty and activates the cleaning process, provided that the water level is sufficient and the temperature/humidity conditions do not interfere with the dust reading.

A water tank is placed on a 3D-printed load-cell structure. The load cell measures the mass of water in the tank through the HX711 module. If the water level is above the required threshold, the pump can be activated. If the water level is too low, the pump is prevented from running, and a “low water level” message is displayed on the LCD.

A YF-S201 flow sensor is placed after the pump to confirm that water is flowing through the tubing. If the pump is activated but no flow is detected, an error condition is generated and the system is shut down to protect the pump and prevent dry operation.

4. Main Components

The main components used in the prototype are listed in the table below.

Component	Quantity	Function
Arduino Uno	1	Main microcontroller used to read sensors and control actuators
LDR sensors	4	Detect light intensity differences for solar tracking
SG90 servo motors	2	Provide dual-axis panel rotation
TCRT5000 infrared reflective sensor	1	Detects dust by measuring reflected infrared light
DHT22 humidity and temperature sensor	1	Measures relative humidity and supports environmental compensation
YF-S201 Flow sensor	1	Detects water flow through the tube after pump

		activation
HX711 module	1	Converts the load cell signal into a digital reading
5kg load cell	1	Measures the water mass in the tank
Submersible water pump	1	Pumps water from the tank to the panel
5V two channel relay module	1	Switches the pump using the Arduino control signal
LCD screen	1	Displays water-level status, temperature, and humidity
Demo solar panel	1	Represents the panel being tracked and cleaned
Breadboard	2	Used for circuit prototyping
Resistors	6	Used for circuit protection and proper functioning
Tubing	1.5 meters	Transfers water from the tank to the panel
Water tank	1	Stores water for the cleaning system
3D printed base and supports	1	Holds the tracking structure and load cell assembly

Table 1. Summary of the system main components

The TCRT5000 is a reflective optical sensor that contains an infrared emitter and a phototransistor detector, making it suitable for detecting changes in reflected light from a nearby surface. The manufacturer lists a peak operating distance of approximately 2.5 mm and an operating range that depends on reflected collector current, so the sensor position above the panel must be kept consistent during testing.

The HX711 was selected because it is a 24-bit analog-to-digital converter designed for weigh-scale and load-cell applications. This allows the very small load-cell signal to be amplified and converted into a digital value that can be read by the Arduino.

The DHT22 was included to measure humidity and temperature. In this project, its humidity and temperature readings is mainly used to help decide whether the dust-sensor reading is reliable enough for cleaning activation. The DHT22 provides a digital signal and operates from a low-voltage DC supply.

The YF-S201 flow sensor was included to verify water movement through the cleaning tube. Its output is a pulse signal proportional to flow rate, and a common datasheet relation is given by frequency in Hz = $7.5Q$, where Q is the flow rate in L/min.

The Arduino Servo library was used for servo control, since it allows Arduino boards to control hobby servo motors.

The LiquidCrystal library was used for the LCD display because it supports common HD44780-compatible character LCD modules. A 16x2 LCD with I2C interface was used.

5. Mechanical Design

The prototype was built around a 3D-printed conical base with a flat top. The first servo motor was placed on the flat top of the conical base. This bottom servo provides rotation around one axis. A rectangular 3D-printed frame was connected to this servo. A second servo motor was mounted orthogonally on the rectangular frame, and this second servo was connected to the panel holding frame. In this way, two perpendicular rotations were achieved, allowing the panel to perform dual-axis tracking.

The demo solar panel was attached to the panel holding frame using glue. The four LDR sensors were mounted on the moving panel itself, not on a stationary external sensor block. Barriers were placed between the LDRs to create shadows when the light source is not aligned with the panel. These barriers improve the sensitivity of the tracking system because small changes in light direction produce differences between opposite LDR readings.

The water tank was placed on a 3D-printed load-cell support structure. This structure was designed so that the load cell could measure the tank weight properly. The submersible pump was placed inside the water tank. Tubing was connected from the pump outlet to the solar panel surface.



Fig1. Bottom section of the solar tracker

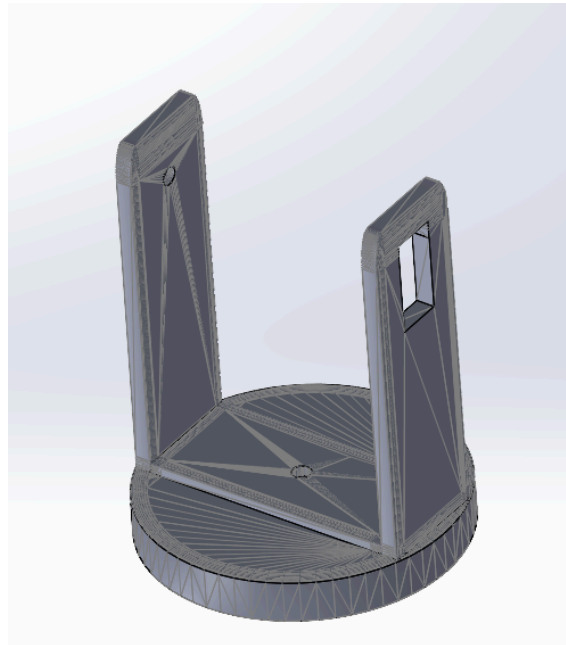


Fig2. Middle section of the solar tracker

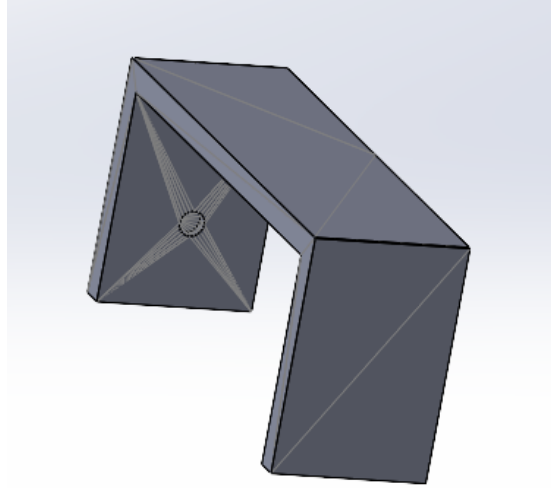


Fig3. Solar panel mount

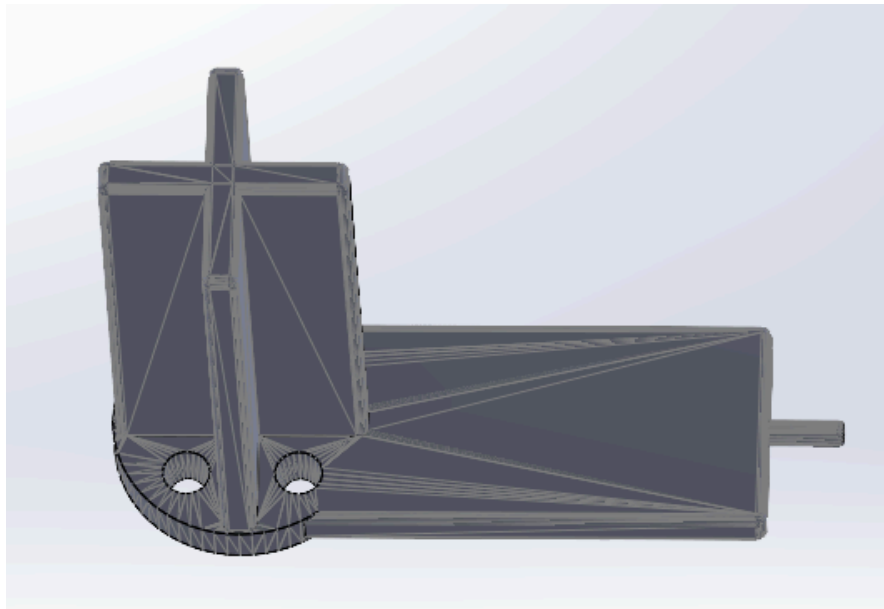


Fig4. LDR mount

6. Electrical and Control Architecture

The Arduino Uno was used as the central controller. The LDR sensors, dust sensor, temperature sensor, humidity sensor, load cell module, and flow sensor provide input signals to the Arduino. The servo motors, relay-controlled pump, and LCD are output devices controlled by the Arduino.

The pump was not powered directly from the Arduino. Instead, a separate battery supply was used for the pump, and the relay module was used as an electrically controlled switch. This separation is important because the pump requires more current than an

Arduino output pin can safely provide. A common ground should be maintained between the Arduino circuit and external supply where required by the module configuration.

The system was assembled on a breadboard. This was suitable for the prototype stage because it allowed the circuit to be modified and corrected during testing.

Sensor	Pin
Top right LDR	A2
Top left LDR	A1
Bottom right LDR	A3
Bottom left LDR	A0
Servo horizontal (azimuth)	D2
Servo vertical (tilt)	D6
Pump regulator module	D8
Temp+humidity	D7
Load cell (HX711)	SCK=D4 DT=D5
Flow sensor	D3
Ir sensor (dust)	D9
I2C	SDA = A4 SCL = A5

Table 2. Table showing the electrical components connections

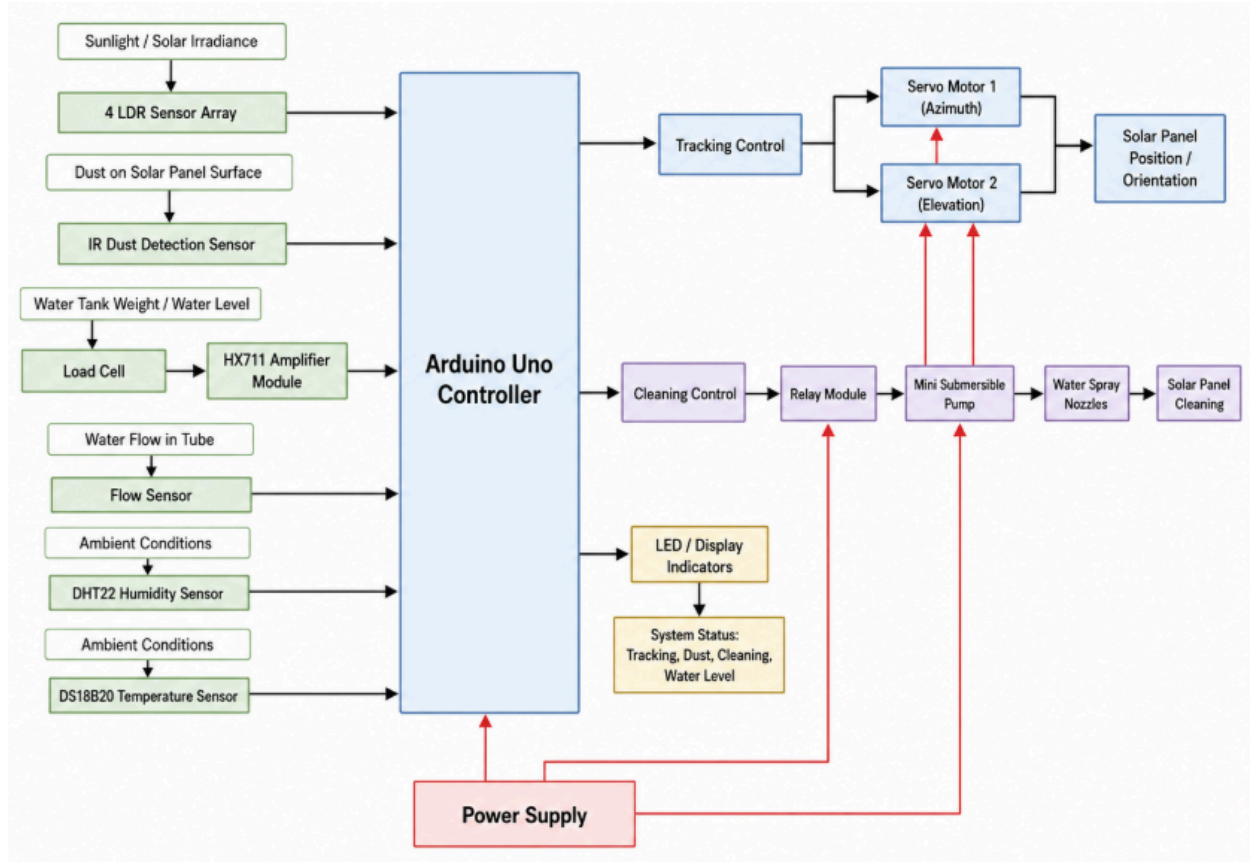


Fig 5. System architecture diagram

7. Solar Tracking Method

The solar tracking system is based on four LDR sensors mounted on the moving panel. The LDRs are arranged as top-left, top-right, bottom-left, and bottom-right sensors. The purpose of the tracking algorithm is to compare the light intensity received by opposite sides of the panel and rotate the panel until the difference becomes small.

The four readings may be defined as:

TL = top-left LDR reading

TR = top-right LDR reading

BL = bottom-left LDR reading

BR = bottom-right LDR reading

The average light intensity on each side is then calculated as:

Top average: $T = (TL + TR) / 2$

Bottom average: $B = (BL + BR) / 2$

Left average: $L = (TL + BL) / 2$

Right average: $R = (TR + BR) / 2$

The vertical tracking error is: $e_{\text{vertical}} = T - B$

The horizontal tracking error is: $e_{\text{horizontal}} = L - R$

If the top average is greater than the bottom average, the panel is rotated upward. If the bottom average is greater than the top average, the panel is rotated downward. Similarly, if the left average is greater than the right average, the panel rotates toward the left, and if the right average is greater, the panel rotates toward the right.

To prevent unnecessary servo jitter, a tolerance band should be used. A starting tolerance is:

LDR tolerance = 50 ADC counts

This means that the servo motor should not move unless the absolute difference between opposite averages is greater than 50 counts. This value corresponds to approximately 4.9% of the full Arduino ADC range, since the Arduino analog input range is from 0 to 1023 counts. This threshold is based on testing where some values resulted in the whole structure vibrating while others increased the response time making the rotation very slow.

8. Dust Detection Method

The dust detection system is based on the TCRT5000 infrared reflective sensor. The sensor emits infrared light toward the panel surface and measures the amount of reflected light. A clean panel surface produces a certain baseline reflection. When dust is added to the surface, the reflection changes. This change is used as an indication that cleaning is required.

Because the exact analog reading depends on the sensor height, panel color, ambient light, and dust type, the dust threshold should be obtained experimentally rather than assumed as a fixed universal value. The following calibration procedure was followed:

1. The clean panel should be measured several times.

2. The average clean reading should be calculated.
3. Dust should be added to the panel surface.
4. The dusty reading should be measured several times.
5. A threshold should be selected between the clean and dusty readings.

The values vary depending on the room and ambient light setting which leads to variation in the threshold and baseline value, this forces calibration on every test run depending on the surrounding factors.

Therefore since there are no fixed numbers, this demonstration is only to show the logic of the process:

S_{clean} = average sensor reading for clean panel

S_{dusty} = average sensor reading for dusty panel

S_{current} = current sensor reading

A simple dust reflection index may be defined as:

$$\text{Dust Index (\%)} = |S_{\text{current}} - S_{\text{clean}}| / S_{\text{clean}} \times 100$$

A starting threshold is Dust Index threshold = 20%

Therefore, cleaning is triggered if Dust Index $\geq$ 20%

This threshold is only for the sake of the demonstration, experimental testing results in different thresholds as previously mentioned.

9. Environmental Condition Check

The DHT22 sensor is used to monitor humidity and temperature. In this prototype, these readings are not only displayed on the LCD but are also used in the cleaning decision.

When both humidity and temperature are high, cleaning is prevented because environmental conditions can affect the reliability of the dust sensor reading. High humidity can change surface reflection, and temperature can affect sensor behavior and water evaporation from the panel.

The following is a demonstration of the decision making logic behind this sub-system:

RH = relative humidity measured by the DHT22

T = temperature measured by the DHT22

If $RH \geq 70\%$ and $T \geq 35^{\circ}\text{C}$, cleaning is disabled.

The values of 70% relative humidity and 35°C are starting values. They are to be adjusted according to testing conditions.

The goal of this sub-system is to prevent any water waste in the cases of unreliable sensor readings.

10. Water-Level Monitoring Using Load Cell and HX711

A 5 kg load cell was used to measure the water mass in the tank. The load cell was placed under the tank using a 3D-printed support structure. The HX711 module was used to read the load-cell output.

The system does not only detect whether water is present or absent. Instead, it estimates the amount of water available in the tank by measuring mass. Since the density of water is approximately 1 kg/L, the water volume in liters can be estimated from the measured water mass in kilograms.

The empty tank is first measured. After measuring, the measured mass corresponds mainly to the water mass: $m_{\text{water}} = m_{\text{measured}} - m_{\text{empty_tank}}$ ($m_{\text{empty_tank}} = 354\text{g}$)

The approximate water volume is: $V_{\text{water}} = m_{\text{water}} / \rho_{\text{water}}$

Using $\rho_{\text{water}} \approx 1 \text{ kg/L}$: $V_{\text{water}} (\text{L}) \approx m_{\text{water}} (\text{kg})$

The required minimum water level is: $V_{\text{water}} > 0.5 \text{ L}$

Therefore, the required minimum water mass is approximately: $m_{\text{water}} > 0.5 \text{ kg}$

The corresponding water weight is: $W_{\text{water}} = m_{\text{water}} \text{ g} \Rightarrow W_{\text{water}} = 4.905 \text{ N}$

Therefore, the cleaning system should only be allowed to run when the measured water mass is greater than 0.5 kg, equivalent to more than approximately 0.5 L of water.

Water-level logic:

Water volume	System Status	Pump Permission
$V > 0.5 \text{ L}$	Sufficient water	Allowed

$0.3 < V < 0.5$	Warning level	Blocked
$V < 0.3$	Low water level	Blocked

Table 3. Water monitoring system conditions

11. Load Cell Calibration

The load cell was calibrated using an external scale. An object with known mass was first measured using the external scale. The same object was then placed on the load cell, and the corresponding raw HX711 reading was recorded. This process allows the Arduino reading to be converted into mass.

x = raw HX711 reading

m = known mass measured by external scale

A linear calibration equation was used: $m = a x + b$, where a is calibration slope and b is calibration offset

More accurate calibration can be done using multiple known masses and fitting a straight line for more reliable results.

Trial	Known mass (g)	Raw data	Calculated mass	Error	Percent error
1	76	65,480	76.14	0.14	0.18%
2	209	179,540	208.77	-0.23	0.11%
3	354	304,760	354.37	0.37	0.1%
4	62	53,220	61.88	-0.12	0.19%

Table 4. Load cell calibration data

The error can be calculated as: $\text{Error} = \text{Measured Value} - \text{Reference Value}$

$\text{Percent Error} = |\text{Measured Value} - \text{Reference Value}| / \text{Reference Value} \times 100$

The average absolute error is: $\text{MAE} = (1/n) \sum |m_{\text{measured}} - m_{\text{reference}}| = (0.14 + 0.23 + 0.37 + 0.12) / 4 = 0.215 \text{ g}$

The root mean square error is: $RMSE = \sqrt{[(1/n) \sum (m_{\text{measured}} - m_{\text{reference}})^2]} = 0.237 \text{ g}$

These error calculations can be used to evaluate the accuracy of the water-level measurement. And are taken into consideration in the decision making circuit.

The load cell calibration showed a close agreement between the known masses and the calculated masses obtained from the HX711 readings. The percent error remained below 1% for all trials in the example dataset, indicating that the load cell calibration was sufficiently accurate for estimating the water mass in the tank. Since the system only needs to determine whether the water level is above or below the 0.5 L threshold, this level of accuracy is acceptable for the prototype

12. Flow Measurement Using the YF-S201 Sensor

The YF-S201 flow sensor was placed after the pump in the water tube. Its purpose is to confirm that water is actually flowing when the pump is turned on. This is important because a pump may be activated electrically while no water is delivered due to an empty tank, blocked tube, loose connection, or pump failure.

The YF-S201 produces pulses as water flows through it. The flow rate can be calculated from the pulse frequency. A common relation for this sensor is: $f = 7.5 Q$.

f = pulse frequency in Hz

Q = flow rate in L/min

The volume of water delivered can also be estimated. Since the sensor is commonly represented as producing approximately 450 pulses per liter: $\text{Volume (L)} = \text{Number of Pulses} / 450$

During operation, if the pump is on but the measured flow rate is approximately zero, the system identifies an error. In this case, the pump is shut down to protect it and to prevent operation without water flow.

After the pump is activated, the flow-sensor pulses are monitored for 3 seconds. If no pulses, or fewer than the experimentally selected minimum number of pulses, are detected during this interval, a no-flow error is generated and the pump is deactivated.

13. Pump and Relay Operation

The cleaning system uses a submersible water pump placed inside the water tank. The pump is powered using separate batteries and is controlled by the Arduino through the relay module. The relay acts as a switch between the pump and its external power supply.

The pump is activated for 5 seconds when all cleaning conditions are satisfied. Cleaning is allowed only when:

1. Dust is detected.
2. Water level is sufficient.
3. Temperature and humidity conditions do not block cleaning.
4. Water flow is detected after pump activation.

14. Complete Control Logic

The complete system operation can be summarized as follows:

1. The Arduino reads the four LDR sensors.
2. The average light intensity on the top, bottom, left, and right sides is calculated.
3. The two servo motors are adjusted until the light difference between opposite sides is within the selected tolerance.
4. The Arduino reads the TCRT5000 dust sensor.
5. The dust index is calculated relative to the clean panel baseline.
6. The DHT22 readings are checked.
7. The load cell reading is converted into water mass and water volume.
8. If dust is detected and water is sufficient, cleaning is allowed.
9. If the water level is low, the pump is blocked, and the LCD displays a low-water message.
10. If cleaning is allowed, the relay activates the pump.
11. The pump sprays water onto the panel through the tube for 5 seconds.
12. The flow sensor checks whether water is flowing.
13. If no flow is detected while the pump is on, the pump is shut down and an error condition is displayed.
14. After cleaning, the pump turns off and the system returns to monitoring mode.

15. Thresholds

Parameter	Threshold Value	Purpose
LDR tracking tolerance	50 ADC counts	Prevents servo jitter and unnecessary movement
Dust Index threshold	20% change from clean baseline	Initiates a cleaning request
Dust confirmation count	3 consecutive readings	Reduces false triggering caused by temporary fluctuations
Low-water warning	Below 0.5 L	Prevents pump operation when insufficient water is available
Pump run time	5 s	Defines the duration of the cleaning cycle
Flow failure threshold	$Q < 1 \text{ L/min}$	Detects blocked tubing, pump failure, or absence of water flow
Flow failure time	3 s	Prevents immediate false error
High humidity threshold	70% RH	Blocks or flags potentially unreliable dust measurements
High temperature threshold	35°C	Prevents cleaning under the selected high-temperature condition

Table 5. Summary of the system threshold

16. Calibration of the Dust Sensor

Since the TCRT5000 dust detection method depends on reflection, the sensor must be calibrated for the specific panel surface and sensor placement used in the prototype. The calibration should be performed under the same lighting conditions and sensor distance

used during the demonstration. The TCRT5000 digital output was used. Its onboard comparator threshold was adjusted using the module potentiometer by testing the sensor with clean and dusty panel conditions. The Arduino then confirmed dust only after three consecutive active digital readings.

This is due to many inconsistencies faced while calibrating the analog output. However the data collected from the calibration is real and was used and tested in the dust detection system.

Panel Condition	Reading 1	Reading 2	Reading 3	Average	Standard Deviation	Dust index	Decision
Clean	762	747	754	754.3	6.0	0.0%	No cleaning
Light dust	675	688	681	681.3	6.5	9.7%	No cleaning
Medium dust	587	602	594	594.3	7.5	21.2%	Cleaning
Heavy dust	430	445	438	437.7	7.5	42.0%	Cleaning

Table 6. Dust calibration data

The dust sensor error can be evaluated using repeated readings. For each condition, the mean and standard deviation should be calculated: Mean reading: $S_{\text{mean}} = (1/n) \sum S_i$
Standard deviation: $\sigma = \sqrt{[(1/(n - 1)) \sum (S_i - S_{\text{mean}})^2]}$

A high standard deviation indicates that the dust reading is unstable. If this occurs, the sensor should be shielded from ambient light, repositioned, or filtered using software averaging. The classification accuracy can also be evaluated: $\text{Accuracy} = \text{Correct Classifications} / \text{Total Tests} \times 100$

Thus in our case $S_{\text{clean}}=754.3$ and the dust index is calculated based on it using the formula: $\text{Dust Index} = (|S_{\text{current}} - S_{\text{clean}}| / S_{\text{clean}}) \times 100$. With a cleaning threshold index of 20% we can use this calculation to confirm or neglect the cleaning order.

The calibration results show that the TCRT5000 sensor reading changed clearly as the dust level increased. The clean panel produced an average reading of 754.3, while the heavy-dust condition produced an average reading of 437.7. Using the clean-panel value

as a baseline, the dust index increased from 0.0% for the clean panel to 42.0% for the heavy-dust condition. A 20% dust-index threshold was therefore considered suitable for triggering the cleaning cycle, since it prevented cleaning under light dust while activating the pump under medium and heavy dust conditions. The calibration curve showed a nearly linear relationship between the dust index and the reference dust level. The Numbers may vary slightly but the general threshold remains intact.

17. Calibration of the LDR Tracking System

The LDR sensors are used in voltage divider circuits. The Arduino reads the output voltage as an analog value between 0 and 1023.

The ADC reading is related to voltage by: $\text{ADC Reading} = V_{\text{out}} / V_{\text{ref}} \times 1023$

For an Arduino Uno using a 5 V reference: $V_{\text{out}} = \text{ADC Reading} / 1023 \times 5$

If the LDR is used in a voltage divider with a fixed resistor, the LDR resistance can be estimated from the measured output voltage. The exact equation depends on whether the LDR is placed above or below the fixed resistor.

For one common arrangement, where the LDR is connected to 5 V and the fixed 10 k Ω resistor is connected to ground: $V_{\text{out}} = V_{\text{cc}} \times 10 \text{ k}\Omega / (R_{\text{LDR}} + 10 \text{ k}\Omega)$

Solving for the LDR resistance: $R_{\text{LDR}} = 10 \text{ k}\Omega \times (V_{\text{cc}} / V_{\text{out}} - 1)$

The system does not require exact lux measurement because it uses relative light differences between LDRs. Therefore, calibration mainly involves checking that all four LDRs give similar readings under uniform lighting. If one LDR gives a much higher or lower value than the others under the same light, a correction factor can be applied.

Corrected reading: $S_{\text{corrected}} = k S_{\text{raw}}$

k = correction factor for each LDR

S_{raw} = raw analog reading

$S_{\text{corrected}}$ = corrected reading

The correction factor can be estimated as: $k = \text{Average reading of all LDRs} / \text{Reading of individual LDR}$

This allows the four LDRs to be balanced before the tracking algorithm is applied.

Condition	TL	TR	BL	BR	Top avg	Bottom avg	Left avg	Right avg	Vertical error	Horizontal Error	Action
Light centered	795	793	797	794	794	795.5	796	793.5	-1.5	2.5	none
Light From left	860	695	848	702	777.5	775	854	698.5	2.5	155.5	Rotate left
Light from right	690	855	705	842	772.5	773.5	697.5	848.5	-1	-151	Rotate right
Light from top	870	855	700	715	862.5	707.5	785	785	+155	0	Titlt up
Light from bottom	705	715	850	865	710	857.5	777.5	790	-147.5	-12.5	Tilt down

Table 7. LDR calibration data

The LDR tracking system was calibrated by exposing all four LDR sensors to the same light intensity and comparing their analog readings. Ideally, all sensors should produce equal readings under uniform lighting; however, small differences occur due to sensor tolerance, resistor variation, wiring, and placement. Therefore, a correction factor was calculated for each LDR by dividing the average reading of all four sensors by the individual sensor reading. This allowed the raw readings to be normalized before being used in the tracking algorithm. After calibration, the tracking response was tested by shining light from different directions. When the light was centered, the difference between opposite sensor pairs remained below the selected tolerance of 50 ADC counts, so no servo movement was required. When the light was moved toward the left, right, top, or bottom side of the panel, the corresponding horizontal or vertical error exceeded the tolerance, and the servo motors rotated the panel toward the stronger light direction. The example results show that the LDR system correctly identified the direction of maximum light intensity and provided suitable input for dual-axis solar tracking.

18. Testing Procedure

The prototype was tested by demonstrating the two main functions: solar tracking and self-cleaning.

For the tracking test, a light source was shined toward the panel from different directions. The four LDR sensors detected the differences in light intensity, and the two servo motors

rotated the panel toward the stronger light direction. The system continuously checked for differences between opposite LDR pairs and corrected the panel position when necessary.

For the cleaning test, dust was placed on the panel surface. The TCRT5000 sensor detected the change in reflected light. Once the dust threshold was reached, the system checked the water level using the load cell. If enough water was available, the pump was activated through the relay and water was sprayed onto the panel through the tube for 5 seconds. The flow sensor confirmed that water was moving through the tube.

The water-level monitoring system was tested by placing the tank on the 3D-printed load-cell structure and checking the system response at different water quantities. When the water was sufficient, the LCD screen indicated safe operation. When the water level was low LCD warnings were activated.

The environmental sensors were tested by displaying temperature and humidity on the LCD and using them as part of the cleaning decision.

19. Results and Discussion

The final prototype was functional. The solar tracking system successfully responded to changes in light direction. When the light source was moved, the LDR readings changed, and the servo motors adjusted the panel position accordingly. This demonstrated the ability of the system to search for a position of maximum light intensity.

The dust detection system also operated successfully. When dust was placed on the panel, the TCRT5000 sensor detected a change in reflected infrared light. Once the selected threshold was reached, the cleaning sequence was activated, provided that the water level and environmental conditions were acceptable.

The cleaning system successfully sprayed water onto the panel through the tube. And produced a cleaner solar panel which yields a better more efficient solar power conversion.

The load cell and HX711 system successfully measured the tank water mass. This allowed the system to determine whether enough water was available before activating the pump. The use of a load cell was more informative than a simple water presence sensor because it allowed the approximate amount of water to be estimated.

The flow sensor successfully verified water movement through the tube. This added a protection feature because the system could detect if the pump was on but no water was flowing. In such a condition, the system was designed to shut down the pump to prevent damage.

Several issues were encountered during development. Initially, the servo motors did not function properly, and the LDR readings were not reliable. These issues were resolved through circuit correction, testing, and mechanical improvement. The structure was improved using 3D printing, which made the final prototype more stable and better aligned. Different servo configurations were also tested before the final motion behavior was achieved.

20. Limitations

Although the prototype was functional, several limitations remain. First, the system was built using a demo solar panel and small servo motors, so it represents the concept rather than a full-scale outdoor solar installation. A real solar panel would be heavier and would require stronger motors, a more rigid frame, weatherproofing, and a more reliable power system.

Second, the dust detection method depends on reflected infrared light. This means that the reading can be affected by sensor distance, panel surface color, ambient light, water droplets, humidity, and temperature. For this reason, calibration is necessary before reliable operation. The TCRT5000 sensor is suitable for the prototype, but a more advanced dust or optical soiling sensor may be required for a full-scale system.

Third, the cleaning mechanism uses only water spraying. Since no brush was used the system may not remove sticky dirt, mud, or particles that strongly adhere to the panel. A future version may require a controlled wiper, brush, air jet, or improved nozzle design.

Fourth, the prototype was assembled on a breadboard. While this is acceptable for testing, it is not ideal for long-term use because wires can loosen and readings can become unstable. A soldered circuit or printed circuit board would improve reliability.

Fifth, the water system depends on the tank capacity and pump strength. If the water level is too low, cleaning cannot occur. If the pump flow rate is weak, the cleaning action may not cover the full panel surface. The nozzle and tube layout should therefore be optimized in future versions.

Finally, the solar tracking system depends on LDR comparison. This method works well in a controlled demonstration, but it may be affected by shadows, reflections, clouds, or nearby light sources. A real outdoor system may require filtering, time delay, or a hybrid control method using both LDR feedback and a sun-position algorithm.

21. Future Improvements

Several improvements can be added in future versions of the project:

- A larger and more rigid tracking frame could be designed for a real solar panel.
- Stronger motors could be used for higher torque and better outdoor performance.
- A better water spray nozzle could be added to distribute water more evenly across the panel.
- A wiper or soft brush could be reintroduced if stronger cleaning is needed.
- The electronics could be transferred from a breadboard to a soldered board or PCB.
- The system could be weatherproofed for outdoor operation.
- IoT monitoring could be added to display panel status, water level, temperature, humidity, and cleaning history remotely.
- A battery and solar charging subsystem could be added to make the system self-powered.
- A more advanced dust detection method could be used to reduce false readings.
- Data logging could be added to compare panel output before and after tracking and cleaning.

22. Conclusion

A smart solar tracking and self-cleaning solar panel prototype was successfully designed and implemented. The project combined multiple sensors and actuators into one integrated Arduino-based system. Four LDR sensors were used to detect light intensity differences and guide a dual-axis tracking mechanism driven by two servo motors. A TCRT5000 infrared reflective sensor was used to detect dust accumulation on the panel surface. When dust was detected, the system checked the water level using a load cell and HX711 module before activating the pump. A flow sensor was used to verify that water was actually moving through the tubing, and the pump was shut down if no flow was detected. Temperature and humidity sensors were also included to display environmental conditions and prevent cleaning when these conditions could affect the dust sensor reading.

The final prototype demonstrated the intended behavior successfully. It was able to track a light source, detect dust, check water availability, activate a water-spraying cleaning cycle, and provide user feedback through an LCD. The project therefore achieved its main objective of creating a prototype solar panel system that improves performance by maximizing light exposure and reducing dust effects.

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